

An irreducible operator algebra constructed from projection shells

MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_atomicShellModel

theorem

Shell partial isometries with rank-one limiting defects can be completed to unitary links that join inequivalent irreducible fibers.

Statement

Let A be a unital complex C^* -algebra and $(\pi_i : A \rightarrow B(H_i))_{i \in I}$ a family of pairwise unitarily inequivalent irreducible unital representations on nonzero complete complex Hilbert spaces. Fix a root $o \in I$ and unit vectors $\xi_i \in H_i$. On $\mathcal{H} = \bigoplus_i H_i$, let e_i be coordinate inclusion, $b_i = e_i \xi_i$, and $\pi = \bigoplus_i \pi_i$. For shell data satisfying the conditions below, there are unitaries $L_i \in B(\mathcal{H})$ such that $L_i b_i = b_o$ and $L_i(P_{i,n} - P_{i,n+1}) = W_{i,n}$ for all i, n . The literal inclusion into $B(\mathcal{H})$ of the norm-closed unital star algebra $T = C^*(\pi(A), \{L_i : i \in I\})$ is irreducible.

Assumptions

For each $i \in I$, let $(U_{i,n})_{n \geq 0}$ and $(V_{i,n})_{n \geq 0}$ be decreasing subspaces of $\mathcal{H}$, with $U_{i,0} = V_{i,0} = \mathcal{H}$. All these subspaces and both intersections admit orthogonal projections. Write $P_{i,n} = P_{U_{i,n}}$, $Q_{i,n} = P_{V_{i,n}}$, and $P_{i,\infty} = P_{\bigcap_n U_{i,n}}$, $Q_{i,\infty} = P_{\bigcap_n V_{i,n}}$. The given bounded shell maps $W_{i,n}$ satisfy $W_{i,n}^* W_{i,n} = P_{i,n} - P_{i,n+1}$ and $W_{i,n} W_{i,n}^* = Q_{i,n} - Q_{i,n+1}$. Their limiting defects are prescribed: $P_{i,\infty} = \theta_{b_i, b_i}$ and $Q_{i,\infty} = \theta_{b_o, b_o}$. Here $\theta_{a,b} x = \langle b, x \rangle a$, with the inner product linear in its second variable. No finite, countable or separable restriction is placed on the family. The rank-one conditions are input hypotheses of this generic theorem.

Conclusion

The same chosen family $(L_i)_i$ simultaneously has the unitary, vector-link and termwise-shell properties, and its concrete generated target acts irreducibly on $\mathcal{H}$. All closed reducing subspaces of that target action are therefore 0 or $\mathcal{H}$.

This theorem does not assert $L_o = 1$, simplicity of T , faithfulness of π , or capture of every irreducible representation of T . Its shell operators $W_{i,n}$ are not the cyclic-assembly map W used in the later capture results. The star on operators is the Hilbert adjoint.

Proof route

For each i , sum the orthogonal shell maps strongly to an operator S_i , then fill its one-dimensional defect by $L_i = S_i + \theta_{b_o, b_i}$. To prove irreducibility, a bounded operator commuting with the target first has scalar diagonal blocks and zero off-diagonal blocks relative to the atomic source. Commuting with the links forces all those scalars to agree.

▼ Proof steps

1. Apply the cited projection-shell rank-one completion theorem separately to each i , with initial vector b_i and final vector b_o . Coordinate inclusion is isometric, so both vectors have norm one. The decreasing normalized flags, the two support equations and the two limiting projection identities are exactly its hypotheses. It supplies S_i with strong convergence $\sum_{n < N} W_{i,n} \rightarrow S_i$ and the unitary completion $L_i = S_i + \theta_{b_o, b_i}$. Choose these witnesses once for all i .
2. The completion theorem also gives $L_i b_i = b_o$ and $L_i(P_{i,n} - P_{i,n+1}) = W_{i,n}$. The rank-one correction vanishes on every initial difference shell because b_i belongs to the limiting intersection. Thus the correction does not change any prescribed shell action.
3. Let $R \in B(\mathcal{H})$ commute with T . Since $\pi(A) \subseteq T$ and $L_i \in T$, it commutes with the atomic source and every link. The cited atomic-commutant theorem applies to the given irreducible, inequivalent fibers: each matrix block $e_j^* R e_i$ intertwines the corresponding actions; the Schur argument makes diagonal blocks scalar, while irreducibility and unitary inequivalence force off-diagonal blocks to vanish.
4. Write $R e_i x = e_i(z_i x)$. Commuting with L_i and evaluating at b_i gives $z_i b_o = z_o b_o$, hence $z_i = z_o$ since $\|b_o\| = 1$. Finite-support vectors are dense in the arbitrary-index Hilbert sum, so continuity yields $R = z_o 1$ on all of $\mathcal{H}$. Finally, the projection onto any closed reducing subspace commutes with T ; being a scalar projection, it is 0 or 1. This is the cited scalar-commutant irreducibility criterion.

Main citations

- The stated existence or structural result —
MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_atomicShellModel
- Rank-one completion of each strongly summed shell family —
ContinuousLinearMap.exists_rankOneCompletion_of_projectionShells
- Atomic commutant made scalar by the vector links —
MathlibAnnex.Analysis.CStarAlgebra.eq_algebraMap_of_atomic_of_links
- Scalar commutant implies irreducibility —
MathlibAnnex.Analysis.CStarAlgebra.StarAlgHom.isIrreducible_of_commutant_eq_algebraMap
- Concrete generated target — MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget
- Literal inclusion of the concrete target —
MathlibAnnex.CStarAlgebra.AtomicConstruction.ambientInclusion

Supporting route explanation

Lean source signature (exact)

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theorem exists_irreducible_atomicShellModel
  (pi : ∀ i, Representation A (H i))
  (hirr : ∀ i, StarAlgHom.IsIrreducible (pi i))
  (hno : ∀ {i j : I}, i ≠ j → ∀ e : H i ≈i[C] H j,
    → StarAlgHom.Intertwines (pi i) (pi j) (e : H i →L[C] H j))
  (o : I) (xi : ∀ i, H i) (hxi : ∀ i, ‖xi i‖ = 1)
  (W : I → ℕ → HilbertSum H →L[C] HilbertSum H)
  (U V : I → ℕ → Submodule C (HilbertSum H))
  [∀ i n, (U i n).HasOrthogonalProjection]
  [∀ i n, (V i n).HasOrthogonalProjection]
  [∀ i, (⊓ n, U i n).HasOrthogonalProjection]
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[∀ i, (⊔ n, V i n).HasOrthogonalProjection]
(hU : ∀ i, Antitone (U i)) (hV : ∀ i, Antitone (V i))
(hU0 : ∀ i, U i 0 = T) (hV0 : ∀ i, V i 0 = T)
(hInitial : ∀ i n, ((W i n)†).comp (W i n) =
  Submodule.projectionShell (U i) n)
(hFinal : ∀ i n, (W i n).comp ((W i n)†) =
  Submodule.projectionShell (V i) n)
(hUinf : ∀ i, (⊔ n, U i n).starProjection =
  InnerProductSpace.rankOne ℂ (coordinateEmbedding i (xi i))
  (coordinateEmbedding i (xi i)))
(hVinf : ∀ i, (⊔ n, V i n).starProjection =
  InnerProductSpace.rankOne ℂ (coordinateEmbedding o (xi o))
  (coordinateEmbedding o (xi o))) :
∃ L : I → HilbertSum H →L[ℂ] HilbertSum H,
  (∀ i, L i ∈ unitary (HilbertSum H →L[ℂ] HilbertSum H)) ∧
  (∀ i, L i (coordinateEmbedding i (xi i)) =
    coordinateEmbedding o (xi o)) ∧
  (∀ i n, (L i).comp (Submodule.projectionShell (U i) n) = W i n) ∧
  Representation.IsIrreducible
  (ambientInclusion (atomicRepresentation pi) L)

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In the source	Mathematical meaning
$A; I; H\ i; \text{HilbertSum } H$	A unital complex C^* -algebra A , any index set I , nonzero complete complex Hilbert fibers H_i , and their square-summable Hilbert direct sum $\mathcal{H} = \bigoplus_i H_i$.
$\text{pi} : \forall i, \text{Representation } A\ (H\ i); \text{hirr}$	The given unital star representations $\pi_i : A \rightarrow B(H_i)$ are irreducible on their nonzero fibers.
$\text{hno} : \forall \{i\ j : I\}, i \neq j \rightarrow \forall e : H\ i \simeq_{i\ i}[\mathbb{C}] H\ j, \neg \dots$	Distinct indices have no intertwining unitary $E : H_i \rightarrow H_j$ satisfying $E\pi_i(a) = \pi_j(a)E$ for all $a \in A$.
$o : I; \text{xi} : \forall i, H\ i; \text{hxi} : \forall i, \ \text{xi}\ i\ = 1$	Fix a root o and unit vectors ξ_i . The coordinate inclusion e_i gives $b_i = e_i\xi_i \in \mathcal{H}$; hence $b_o = e_o\xi_o$.
$W : I \rightarrow \mathbb{N} \rightarrow \text{HilbertSum } H \rightarrow L[\mathbb{C}] \text{ HilbertSum } H$	The input operators $W_{i,n} : \mathcal{H} \rightarrow \mathcal{H}$ are bounded complex-linear shell maps.
$U\ V; \text{Submodule } \mathbb{C}\ (\dots)$	For each index i and natural n , subspaces $U_{i,n}, V_{i,n}$ of $\mathcal{H}$. Write $P_{i,n}$ and $Q_{i,n}$ for their orthogonal projections.
$[\forall i\ n, (U\ i\ n).HasOrthogonalProjection]; [\forall i\ n, (V\ i\ n).HasOrthogonalProjection]$	Every subspace in both flags admits an orthogonal projection, as required separately for each i, n .
$[\forall i, (\bigcap n, U\ i\ n).HasOrthogonalProjection]; [\forall i, (\bigcap n, V\ i\ n).HasOrthogonalProjection]$	The two intersections $\bigcap_n U_{i,n}$ and $\bigcap_n V_{i,n}$ also admit orthogonal projections, written $P_{i,\infty}, Q_{i,\infty}$.
$\text{hU} : \forall i, \text{Antitone } (U\ i)$	For every i the initial flag decreases: $m \leq n$ implies $U_{i,n} \subseteq U_{i,m}$.

In the source	Mathematical meaning
$\text{hV} : \forall i, \text{Antitone } (V\ i)$	For every i the final flag decreases: $m \leq n$ implies $V_{i,n} \subseteq V_{i,m}$.
$\text{hU0} : \forall i, U\ i\ 0 = \top$	The initial flag begins with $U_{i,0} = \mathcal{H}$.
$\text{hV0} : \forall i, V\ i\ 0 = \top$	The final flag begins with $V_{i,0} = \mathcal{H}$.
$\text{hInitial} : \forall i\ n, ((W\ i\ n)^\dagger).\text{comp } (W\ i\ n) = \text{Submodule.projectionShell } (U\ i)\ n$	For every i, n , $W_{i,n}^* W_{i,n} = P_{i,n} - P_{i,n+1}$. Postfix $\dagger$ is the Hilbert adjoint and .comp composes operators rightmost first.
$\text{hFinal} : \forall i\ n, (W\ i\ n).\text{comp } ((W\ i\ n)^\dagger) = \text{Submodule.projectionShell } (V\ i)\ n$	For every i, n , $W_{i,n} W_{i,n}^* = Q_{i,n} - Q_{i,n+1}$, the final support of the same shell map.
$\text{hUinf} : \forall i, (\prod n, U\ i\ n).\text{starProjection} = \text{InnerProductSpace.rankOne } \mathbb{C} \dots$	The initial intersection projection is $P_{i,\infty} = \theta_{b_i, b_i}$, with $\theta_{a,b}x = \langle b, x \rangle a$ and inner product linear in its second entry.
$\text{hVinf} : \forall i, (\prod n, V\ i\ n).\text{starProjection} = \text{InnerProductSpace.rankOne } \mathbb{C} \dots$	The final intersection projection is $Q_{i,\infty} = \theta_{b_o, b_o}$, with the same root vector b_o for every i . Both rank-one identities are given hypotheses, not automatic properties of arbitrary flags.
$\exists L : I \rightarrow \text{HilbertSum } H \rightarrow L[\mathbb{C}] \text{ HilbertSum } H$	There is one family of bounded operators L_i with all four following outputs simultaneously.
$\forall i, L\ i \in \text{unitary } (\dots)$	Every L_i is a unitary on $\mathcal{H}$.
$\forall i, L\ i (\text{coordinateEmbedding } i\ (xi\ i)) = \text{coordinateEmbedding } o\ (xi\ o)$	Every L_i sends its coordinate vector b_i exactly to the same root vector b_o .
$\forall i\ n, (L\ i).\text{comp } (\text{Submodule.projectionShell } (U\ i)\ n) = W\ i\ n$	For every i, n , $L_i(P_{i,n} - P_{i,n+1}) = W_{i,n}$ as operators on $\mathcal{H}$.
$\text{Representation.IsIrreducible } (\text{ambientInclusion } (\text{atomicRepresentation } \pi)\ L)$	The inclusion of $T = C^*(\pi(A), \{L_i : i \in I\}) \subseteq B(\mathcal{H})$ acts irreducibly, where $\pi = \bigoplus_i \pi_i$. The target is generated with this same family L .

Further source notes: The operators S and the commutant calculation occur in the linked proof, not as extra inputs in the signature.

Earlier published Card and PDF

▼ Exact content identity

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